

Re-examining User Burden in Human-AI Interaction: Focusing on a Domain-Specific Approach

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ABSTRACT

In my thesis, I revisit, specialize, and expand the concept of ‘user burden’ in three different contexts. In the first study, I explore what user burdens occur when deleting unused apps. To do so, I have conducted in-depth interviews, designed questionnaires, and performed scenario-based experiments. In my second study, I designed and developed a conversational agent that documents and reports cases of sexual assault survivors to the police on behalf of the survivors. To discover survivors’ burdens and find solutions to mitigate them, I conducted in-depth interviews and participatory design sessions with sexual assault survivors, as well as diverse stakeholders (e.g., police officers, counselors). In my third study, I investigated why employees resist algorithmic evaluations in workplaces and how to mitigate these burdens. The goal of participating in this doctoral consortium is to share the three lines of research for my thesis with researchers and professors and gain diverse ideas and feedback from the HCI community to better synthesize my works.

CCS CONCEPTS

• **Human-centered computing (HCC);** • **Human-computer interaction (HCI);** • **HCI design and evaluation methods;** • **User studies;**

KEYWORDS

User Burden, Human-AI Interaction, Conversational Agents, Algorithmic Management, Future of Work, Metaverse Workspace, Remote Work

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1 RESEARCH OVERVIEW AND GOALS

I am an eight-year doctoral candidate in the Department of Communication at Seoul National University, advised by Dr. Joonhwan Lee, who directs the Human-Computer Interaction + Design Laboratory.

With an academic background in Management Information Systems (MIS) under business administration (bachelor’s and master’s degrees), my primary research domain is the ‘future of work’ and ‘algorithmic management.’ Specifically, I have strong interests in discovering and solving sociotechnical problems of designing and adopting emerging technologies (i.e., AI and Metaverse platforms) in organizational settings. My past research has focused on investigating opportunities, challenges, and design implications at both individual and organizational levels when adopting 1) Metaverse workspace at a large scale [3] and 2) AI-based human resource (HR) management/evaluation systems in enterprises [1, 2]. In this line of my research stream, three full first-authored papers have been published at CHI 21, 22, 23, and one paper (also full first-authored) has been conditionally accepted at CHI 24.

Additionally, with my Ph.D. background in Communication, I aim to discover individual users’ psychological burdens and challenges when conversing with AI (e.g., conversational agents [CAs] and ChatGPT) and find solutions – e.g., not only individual-level UI or UX design but also macroscopic solutions such as social systems and policy design – to mitigate them. More specifically, I led a government-based research project as a project manager (first-year) – designing and developing a conversational agent (CA) that documents and reports a case of sexual assault to the police on behalf of sexual assault survivors. During this project, I conducted several studies that examine 1) what psychological burdens emerge when a survivor discloses their story to a CA [7] and 2) what communication/interaction breakdowns occur when such a CA is adopted into the broader social (police) system as an active social actor [5]. Each first-authored paper has been published at CHI 20/21 [6, 7] and DIS 21 [5]. Most recently, I have expanded my research interest under the broader theme of uncovering sociotechnical issues in human-AI interaction. One full (first-authored) paper on discovering students’ challenges in conversing with ChatGPT for learning in higher education has also been conditionally accepted at CHI 24.

I plan to complete my Ph.D. program next semester, weaving my ongoing and published studies into an impactful thesis. I will dedicate myself with passion and vision to producing new knowledge for the human-computer interaction scholarship, leveraging my strengths in openness, adventurousness, and interdisciplinarity in research.

2 THESIS OVERVIEW AND GOALS

To improve computing systems, the HCI community has endeavored to develop ways of evaluating them along various dimensions, including usability, usefulness, and affective experience [8]. Also, relevant theories (e.g., uses and gratification theory and technology acceptance model) delve into users’ specific needs or perceived

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usefulness toward a certain system by emphasizing the functionality and/or positive aspects of a system. Although researchers and practitioners put tremendous effort into advancing a system's functionality, some users still do not adopt or continue to use the system. This may be because users' system use is attributed not only to system functionality (e.g., usability/needs) but also to subjective factors, which may be dependent on individual differences. Thus, traditional measures such as usability and usefulness alone cannot fully explain either system adoption or the willingness to continue using the system.

To overcome such limitations of the measures, Suh et al. [8] presented user burden, which refers to "the negative impact that computing systems might place on users." Although user burden incorporates two of the most representative concepts in HCI – usability and user experience –, it distinguishes itself from traditional concepts in that it can also include other psychological and subjective dimensions influenced by individual characteristics [8]. Suh et al. [8] explain that, although a system could benefit people greatly in terms of system functionality, some users might not adopt the system in the first place or stop using it. For example, some users might not want to risk their privacy (e.g., disclosing personal information to CAs), while others are willing to give their information for the consequent benefits. This shows that people might use useful functionalities when necessary, but it could also have a negative influence on their lives.

However, the framework has limitations for the following reasons. First, the concept of burden deals with a meta concept as an umbrella term. Although user burden is defined as "the negative impact that computing systems might place on users [8]," it is still difficult to differentiate it from the dictionary term "burden" or "psychological barriers" which are often used in other domains (e.g., psychology and communication). Second, the authors pursued universality for scale development. They allowed "extremely general users (i.e., MTurk users)" to evaluate the user burden scales regarding "extremely general systems (i.e., Gmail, Facebook, and Netflix)." Therefore, the framework cannot suit a unique context in which specific users (e.g., vulnerable populations) interact with a system. Third, the scale could be inappropriate to measure a system with an intangible interface (e.g., voice-user interface and algorithm) because the computing systems that respondents chose to assess the scales in Suh et al. [8]'s experiment were all equipped with a tangible interface (e.g., Facebook's website). Finally, user burden scales were developed and validated based on users' experiences of abandoning a system that was once adopted and used. Thus, the user burden scales cannot either catch or explain users' burdens that occur "before" adopting and using the system.

In this thesis, I aim to revisit the user burden framework and examine three different domains in which critical real-world problems emerge and, thus, should be solved. More specifically, I investigate three different contexts in which people feel reluctant to act in certain ways: S1) smartphone users' unused app deletion burdens, S2) sexual assault survivors' burdens of self-disclosure, and S3) employees' burdens of adopting an algorithmic decision on work performance evaluation. Although the three domains are distinctive by nature, they have one thing in common; users necessarily go through an unknown/uncertain situation after making a decision and/or interacting with a system.

In the case of S1, even though mobile apps are fully owned by the user, the potential risks of deleting apps could be vague, such as not being aware of future data loss (e.g., what data is stored inside the unused app). Similarly, while sexual assault survivors might feel more comfortable disclosing their story to a machine rather than a human, they could still be suspicious of whether humans exist behind the machine (S2). For instance, explanations about the CA – who owns and/or has access to the CA, to what degree one's privacy can be protected, etc. – are opaque. Finally, in any aspect, interacting with AI as a job performance evaluator is a new experience for people in workplaces (S3). In this situation, a machine is no longer an assistive tool but, contrarily, evaluates a human in opposite power dynamics (i.e., a machine in a higher position). This unexpected situation without proper communication makes their work environment uncertain and risky.

The three different domains were selected based on the interaction level mainly, as well as by including stakeholders, interfaces, tasks, and machine automation categories. Among them, the first is the domain of personal information management (i.e., unused mobile application deletion) in which an individual should make decisions to delete unused apps or not. This interaction occurs between a mobile app owner and the app(s), meaning the decision is solely made by the individual. The second is based on a more interactional level because survivors' self-disclosure inevitably occurs through human-human communication in the traditional case-reporting system. Although various stakeholders could be involved in the case-reporting process, most of the survivors' self-disclosure is based on dyadic communication (i.e., police-survivor, counselor-survivor, etc.). Finally, accepting AI at workplaces as a performance evaluator should be viewed through a broader lens. This is because a variety of stakeholders (e.g., manager, HR team, peer, etc.) could directly engage in work performance evaluation; at the same time, its impact could significantly harm employees' family, work environment, relationships with colleagues, etc. Also, it could be affected by national policy, law, and ethics related to the labor market at the societal level. By both specializing and incorporating the three different domains, this thesis could cover a variety of burdens that occur and could be applied to ranging from individual to organizational/societal levels. Also, it enables a better understanding of burdens derived by unique characteristics of interface, stakeholders, and task differences.

Applying the concept of user burden to understand people's reluctance in such various contexts is important in that it could demonstrate how well the concept could explain some of the most problematic situations that arise in our everyday lives by revisiting and extending prior work. Unfortunately, the negative aspects of a system (i.e., user burden) have not been examined as much as usability, which is the most well-known and used assessment standard in HCI. However, user burden still has room for improvements, applications, and extensions for a better understanding of users' system adoption and abandonment. Also, it is important for researchers to understand how users experience various types of burdens in diverse contexts in the process of communicating with a system. By examining and applying user burden to three different contexts, my thesis contributes to 1) framework development for each context and 2) design efforts to alleviate burdens emerging in each system design.

To sum up, this study aims to investigate what burdens people have in three different contexts and explore how to reduce them: 1) smartphone users' unused app deletion burdens, 2) sexual assault survivors' burdens of self-disclosure, and 3) employees' burdens of adopting algorithmic decisions on work performance evaluation. In the following section, I will explain each study in detail.

3 THESIS PROGRESS TO DATE

3.1 Burden Study 1: Understanding Smartphone Users' Hesitation in Deleting Unused Apps

I conducted three sets of studies—1) interviews, 2) questionnaire design and principal component analysis (PCA), and 3) scenario-based validation study—to understand whether smartphones have difficulties in deleting their unused apps and, if so, why users hesitate to delete them. In the first study, my co-authors and I interviewed ten smartphone users, conducting thematic analysis to broadly uncover users' psychological burdens in removing unused apps. As a result, we found that smartphone users feel time, mental, emotional, privacy, and information, financial burdens in deleting unused apps. The findings of our preliminary study have been published in the poster presentation at the MobileHCI conference [4].

I have conducted the other two studies to 1) develop app deletion burden scales and 2) discover different burdens emerge when a user deletes each unused app. Based on the interview results and reviews of previous research, I developed six constructs of app deletion burden scales and validated them through PCA. Also, I conducted a between-subjects online experiment through Amazon Mechanical Turk by using our burden scales with ten app deletion scenarios. Based on the initial exploration and construct validity using PCA, I refined the initial proposed categories into a model consisting of six constructs. A definition of each of the final constructs is as follows – that is, *Time Burden*: It requires a significant amount of time. *Mental & Emotional Burden*: It requires a significant amount of attention and concentration, or makes the users feel uncomfortable or worried. *Paid-app Burden*: It risks financial loss because it is a paid app. *Rewards Burden*: It risks financial loss because it has discount coupons and mileage inside the app. *Data Burden*: It risks users' personal information including ID, password, health data, image, note, etc. *Social Burden*: It risks social loss because there are social networks which are potentially beneficial–.

3.2 Burden Study 2: Understanding Sexual Assault Survivors' Burdens of Self-Disclosure

In my doctoral studies, I designed and developed a conversational agent (CA) for sexual assault survivors by actively engaging all stages of the research and development processes [7]. The aim of the study [7] was to identify psychological burdens that arise when survivors' interact with AI or humans (e.g., police officers) and to mitigate the burdens through survivor-centered solutions. Overall, I pursued a design thinking approach so that survivors' real interactions with a tangible AI artifact could give shape to feasible AI design without overestimating AI's capabilities. My study showed that many survivors had experienced various burdens (i.e., time, financial, availability, mental, emotional, privacy & security, social,

physical, data) when interacting with humans to report their case and disclose their stories. While many such burdens were mitigated by interacting with a CA (and seventeen survivors' preferred reporting their case to the CA over police officers), I found that critical CA-bound burdens still exist (e.g., privacy & security and data burdens) and are contextualized within uniquely high-risk and high stress situations (e.g., intimate partner and domestic violence). For example, while relatively affluent groups (e.g., consumer IoT users) may not want too much privacy & security interfaces because they seek convenience and efficiency, survivors are willing to exert significant effort on them, which reflects the particularly risky situations they face. Thus, one of the CA features designed by the survivors participating in the study was multi-stage authentication.

3.3 Burden Study 3: Understanding Employees' Reluctance to Adopt Algorithmic Decision on Work Performance Evaluation

As AI permeates our lives and living spaces, it also impacts workers in workplaces where AI performs HR tasks, such as screening resumes, managing workers, and even evaluating work performance. The case of Amazon tracking and firing its employees through AI without human intervention motivated me to explore why the workers resisted AI's decision-making and how to mitigate workers' burdens towards AI. Because the use case of AI in HR is too new to recruit workers who have been fired by AI, I designed a unique scenario-based interview method that enabled participants who had not yet experienced AI in their workplace to develop imaginative empathy with the situation [1]. Consequently, I discovered six types of burdens (i.e., emotional [= uncanny valley and inhumanity], mental, bias, manipulation, privacy, and social burden) workers feel toward AI and further identified workers' demands for transparency, interpretability, and human intervention in AI's decision-making. My research helped provide both researchers and practitioners insight into workers' burdens and solutions, which otherwise would have been discovered through harsh trial and error. As one of the earliest studies to scrutinize AI's application to the HR context, this study was published at CHI 21 and selected as the outstanding international paper of the year at the HCI Korea conference in 2022.

4 NEXT STEPS

As the next steps for my thesis, I will synthesize different types of user burdens discovered in the three specific contexts described above. This series of processes will uncover common burdens that emerge in general contexts and identify unique burdens created in particular contexts. Simultaneously, I will develop a framework that explains how individual, interactional/organizational, and societal-level burdens emerge and identify different approaches to mitigating these burdens. As the next steps for my broader research stream, I will continue to design, develop, and test algorithmic management/evaluation systems that align with various stakeholders' needs and values. Thus, my ultimate research goal is to build a system that benefits multiple stakeholders – HR managers, employees, and employers – without causing harm or burdens on individual system users.

5 BENEFITS AND CONTRIBUTION STATEMENT

Although all three studies have been completed, and some of them have already been published at CHI (full papers), combining them into one impactful thesis is still challenging. Therefore, I expect to gain ideas and feedback from prestigious professors and promising students worldwide. Specifically, since my thesis encompasses three completely different research topics under the theme of user burden, diverse viewpoints from researchers with various majors (e.g., Business, Computer Science, AI, Psychology) or research interests (e.g., Science and Technology [STS], marginalized user studies, mobile devices, etc.) would be extremely valuable.

At the same time, I am confident that I can share various research and industrial project experiences of mine with other students and professors participating in the doctoral consortium. I have experience successfully conducting studies with vulnerable populations, including over twenty sexual assault survivors, as well as with police officers, counselors, and legal authorities when developing a conversational agent for survivors. I have also closely collaborated with enterprises, including CEOs, business leaders, AI experts, employees, and HR managers while conducting research on Metaverse workspace and algorithmic management/evaluation. I have published five full first-authored papers at CHI (4) and DIS (1), and two other first-authored papers have been conditionally accepted at CHI 24. I believe that my research experiences and interdisciplinary background will be helpful for other participants to gain multifaceted and creative viewpoints.

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